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3D spectral analysis in the VNIR–SWIR spectral region as a tool for soil classification

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ABSTRACT

Visible, near-infrared and shortwave-infrared (VNIR–SWIR) spectroscopy has proven to be an efficient, rapid and low-cost method for soil spectral analysis that can improve on the results obtained from today's traditional methods of conducting soil surveys. Nonetheless, this tool is used mostly in the laboratory and at surface level. The main objective of this paper is to develop a new optical method for characterizing soil profiles, towards improving the efficiency and accuracy of the traditional soil survey. We used airborne hyperspectral data from the AisaFENIX sensor for surface classification and ASD spectral measurements of soil samples for subsurface analysis. A total of 643 soil samples were extracted from 48 cores, each core representing a soil profile. All samples were air-dried, crushed and sieved, and then analyzed by ASD spectrometer under laboratory conditions. Clay content was also measured to provide additional information. The 3D spectral data were analyzed using *SAM* algorithm, spectral gradient (m), k-means clustering and gley horizon parameter (G) to classify soils and distinguish between soil horizons in each core. The results suggest that these parameters can provide satisfactory results both from laboratory measurements and hyperspectral remote sensing data (R^2 = 0.81 for clay content and R^2 = 0.78 for gleying conditions) in order to distinguish between the soil horizons using 3D spectral information. Moreover, the method is satisfactory for obtaining soil types from 3D spectral sensing as well as evaluating the catena development and other spatial soil distributions.

1. Introduction

The soil survey, where land is evaluated for ecological purposes and food production, is the most essential part of the soil-mapping process but is still considered the most expensive part of soil research. Soil mapping is challenging and a thought-provoking aspect of the soil science discipline; it requires professional knowledge on fundamental processes occurring in the soil system—its formation and responses to development and change (Hartemink et al., 2012). Today, great effort is being invested in digital soil mapping (DSM) to represent soils and their properties on a large scale (Ashtekar and Owens, 2013). Digital mapping relies on field, laboratory, and remote-sensing observations which are integrated with quantitative methods to infer spatial patterns of soils across various spatial and temporal scales (Grunwald, 2010).

Soils are classified by common properties for the purpose of systematizing knowledge about soil and the processes that control similarities and dissimilarities within a group (Ashman and Puri, 2002). Early classification systems grouped soils according to the conditions forming each soil type, whereas modern systems tend to place more emphasis on observable and measurable properties. Full detailed discussions on the development of modern soil-mapping methods can be found in the literature (Behrens et al., 2005; McBratney et al., 2000, 2003; Scull et al., 2003a, 2003b; Shi et al., 2009; Yang et al., 2011). To understand soil distribution across the landscape, we need to include the catenary concept which describes the lateral variation of soils where a slope exists (Young, 1980). Despite the fact that certain soils are formed under the same environmental conditions, they may differ along the catena. While in the upper part of the catena, the entire soil profile remains above the permanent water table, the lower part may be affected by it, causing gleying and salinization processes (Ashman and Puri, 2002; Singer, 2007; Young, 1980). Gleying conditions occur in red sandy soils (commonly known as “Nazaz”, in the Israel soil classification or “Aqualf” using the USDA soil taxonomy) that include at least one horizon that exhibits pseudogley (less commonly gley) features such as mottling (Singer, 2007). These soils are formed by migration and accumulation of clay from upper horizons to lower strata (e.g., eluviation/illuviation) and its presence may impair subsurface ventilation and drainage. The clay-particle composition in the soil profile

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differs in SiO_2, Al_2O_3, and Fe_2O_3 ratios (Ravikovitch, 1935), with lower SiO_2/R_2O_3 ratios in the Nazaz horizons (Ravikovitch, 1992). Thus, identification of these horizons is important not only for the classification of soil entities but also for farmers.

To apply any kind of soil-mapping activity, special attention has to be paid to the soil sampling, which needs to spatially represent the study area in 3D. Many soil-survey strategies exist, differing mostly in the spatial distribution of their sampling. Choice of the proper method relies on several factors, such as a prior knowledge of the field, elevation changes, and information on the soil-forming factors along with the spatial resolution required for the survey.

Ellis and Mellor (1995) suggested two principal types of survey methods: ground survey and aerial survey. One of the most common in the former category is the grid method. This is a systematic technique that divides the area into small grid cells; soil samples are then collected from the center of each grid cell and the accuracy of the map is determined by the density of the grid cells. For high accuracy, the distance between grid cells is minimized. Since it is a costly and time-consuming method, especially with small grid sizes, it is more accurate than other methods at describing the true distribution of different soil types (Soil Survey Staff, 2014). Although Nanni et al. (2011) found that it is possible to create reliable maps of organic matter and clay even when reducing the sampling density. Another ground-survey strategy is termed "free survey" or "landscape-directed survey". Surveyors use this method when there is preliminary knowledge of the area (e.g. from aerial photography), which enables them to target the sampling using visual surface patterns (color and geometry). In this method, no plot grid is needed, potentially reducing time and number of samples (Ashman and Puri, 2002).

Conversely, modern aerial surveys use surface information obtained from a hyperspectral sensor which records electromagnetic radiation using a panchromatic, multispectral, or hyperspectral sensors (Ben-Dor, 2002). This method can cover large areas in a short time and can help the surveyors target the survey as per the "landscape-directed" method (Buringh, 1960; Burrough et al., 1997).

During the last two decades, much attention has been devoted in many scientific fields to capturing quantitative information from soil-reflectance data (Ben-Dor, 2002; Ben-Dor et al., 2009; Ben-Dor and Banin, 1995; Chang et al., 2001; Dunn et al., 2002; Gomez et al., 2008; Islam et al., 2003; Reeves et al., 2001; Viscarra Rossel et al., 2009; Viscarra Rossel and Webster, 2012; Volkan Bilgili et al., 2010; Zornoza et al., 2008). To that end, the reflectance radiation across the visible near and short infrared (VNIR–SWIR) region (400–2500 nm) is modeled against constituents evaluated by wet chemistry methods with the use of various statistical approaches. The basic idea of this approach is that soil spectral behavior is controlled by several spectrally active soil properties (chromophores) that additively compose the spectrum. Many researchers working on this approach have demonstrated that the optical method is capable of providing low-cost, rapid and reliable products compared to the intensive laboratory work (Arslan et al., 2014; Christy, 2008; Cohen et al., 2005; Dunn et al., 2002; Janik et al., 2009; Reeves and Smith, 2009; Viscarra Rossel et al., 2009; Viscarra Rossel and Webster, 2012; Volkan Bilgili et al., 2010; Zornoza et al., 2008). A number of studies have shown the potential of VNIR–SWIR data analysis using principal component analysis (Odlare et al., 2005; Rizzo and Demattê, 2014; Verheyen et al., 2001) and fuzzy clustering algorithms (Bezdek et al., 1984; De Gruijter et al., 1997; McBratney and de Gruijter, 1992; Odeh et al., 1990; Odgers et al., 2011a, 2011b; Verheyen et al., 2001). Other methods, such as increasing sampling density (Wetterlind et al., 2010) and GIS techniques for modeling soil-generating factors (Karnieli et al., 1998), have proven to be reliable for soil classification based on the relationship between soil properties and soil types. It is only in recent years that the multi-depth approach has been used for soil classification. Vasques et al. (2014) constructed a pseudo multi-depth soil spectrum which provided a novel framework for distinguishing between different soil types with an overall high agreement rates based on the VNIR spectra.

In general, a soil proxy analysis using spectral information can be applied after validation with an unknown sample, serving as a rapid and objective method to learn about the soil. This is essentially based on the fact that soil reflectance is easy to obtain and may provide information about the soil rapidly and simultaneously. Nevertheless, despite its potential and except for a small number of studies (Ben-Dor et al., 2008; Waiser et al., 2007), soil spectroscopy has not yet been used to detail soil entities in the 3D domain, and its potential has not yet been exploited for DSM using the 3D view, which provides essential information for soil classification.

The main assumption of this study is that soil classification is determined by the physical (grain size, morphology, specific surface area) and chemical (clay content, organic matter, mineralogy) compositions in a 3D space. Hence, these physio-chemical properties affect the soil's spectral fingerprint and can thus be used to perform a rapid classification of surface and subsurface samples. The main objective of this work is to develop and implement a new optical method to classify soils based on spectral information in order to improve results obtained from traditional survey. We anticipate that such an approach will bring the soil classification field into the modern era and position it well in the DSM field.

2. Materials and methods

2.1. The study area

The study area consists of agricultural fields in the Ha-Sharon plain, Israel (Fig. 1a) located 10 km northeast of the city of Netanya, at 32°21′23.15″ N and 34°57′28.24″ E. The area is Cas in the Köppen–-Geiger climate-classification system (Kottek et al., 2006) with annual precipitation of 570 mm and a dry summer. Land use is dominated by orchards of citrus fruit and avocado planted a few years before the research was conducted. Fig. 1a provides an aerial photo (1:10,000) of the study area with polygons of the two major soil types in this region, modified from a governmental map website (www.govmap.gov.il). Fig. 1b is a local elevation map with the corresponding west–east cross-section (Fig. 1c), which demonstrates the local catena.

The area was chosen due to the presence of the two soil types: (i) Brown-red sandy soil (Hamra) which correlates well with Rhodoxeralfs (Fig. 1a, polygon 1) accompanied with the Nazaz soil subgroup – Aquic Rhodoxeralf, and (ii) Grumusol – Haploxererts (sometimes classified as Xerofluvents) according to the USDA (Fig. 1a, polygon 2) (Singer, 2007; Soil Survey Staff, 2014). The eastern side of the catena is influenced by the Alexander River, which deposits alluvial fragments on the riverbank, mostly portions of clay and silt which form brown alluvial Grumusol. The western side is elevated 10–20 m above the riverbank (Fig. 1b) and is formed on top of calcareous sandstone "Kurkar".

2.2. The airborne data

The airborne data were acquired using Specim's AisaFENIX sensor which covers the VNIR–SWIR spectral regions (380–2500 nm) in a single continuous image on 17 Sep 2013 at 08:00 UTC. The instrument has excellent sampling intervals with 620 spectral bands (448 were used after pre-processing), FWHM of 3.5–5.5 nm, FOV of 32.3° and high signal-to-noise ratio SNR (600–1000). The flight altitude was 700 m above ground level, resulting in a spatial resolution of 1 m. The flight lines covered a total area of about 0.75 km^2. The Supervised Vicarious Calibration (SVC) method (Brook and Ben-Dor, 2011) has been used for radiometric calibration and the atmospheric effects were removed using the Atmospheric CORrection Now (ACORN) (Imaging and Geophysics, 2001) to obtain the surface reflectance values for further analysis. In order to focus only on the soil coverage, all other surface features (man-made structures, roads, water bodies and vegetation) were masked out using the Spectral Angle Mapper (SAM) algorithm.

2.3. Sampling design

In this study, both sampling methods were chosen to enjoy the advantages of each method: the grid method for high-resolution soil sampling and the landscape-directed method, using the hyperspectral aerial data, for large area coverage. Samples were collected in two field campaigns, on 22 and 29 Sep 2013, using a motorized drilling auger dual tube DT-22 to collect undisturbed continuous soil cores from surface level, down to 2 m depth. A total of 48 cores were taken, out of which 35 by the grid method and 13 by the landscape-directed method for further laboratory measurements and analysis.

2.4. Laboratory measurements

All cores were taken to the Remote Sensing Laboratory and gently cut at tube length to obtain two parts, which were exposed to the atmosphere and air-dried. After drying, each core was divided into 15 soil samples for high sampling resolution: 10 samples were taken from the first m (10 cm each), and 5 samples were taken from 1 to 2 m depth (20 cm each), resulting in a total of 643 dry soil samples. After their categorization, samples were individually crushed and sieved to 2 mm before taking the spectral measurements.

The spectrum of each soil sample was measured with the portable Analytical Spectral Devices (ASD) spectrometer Pro FR in the laboratory. The spectrometer has three detectors with 1-nm spectral resolution that supply 2151 bands in VNIR (350–1000 nm), SWIR1 (1001–1800 nm) and SWIR2 (1801–2500 nm). All samples were measured using a contact probe equipped with a halogen bulb and Spectralon® panel as a white reference which was performed every 15 min in order to benchmark spectral consistency of the measurements. Each sample was measured three times by the same exact protocol and the average spectrum was calculated. For data analysis, only the 400–2450 nm spectral range was used, where a high signal-to-noise ratio exists. In addition, the percentage of clay was measured with a laser diffraction particle-size analyzer (Mastersizer 2000).

2.5. Spectral analysis

In most cases, soil mapping starts with aerial photographs and satellite images to draw boundaries between the soil units by using color and structural changes (Birkeland, 1999). The most common decision-making tool for surveyors is the RGB photograph, which is used to collect preliminary information about the area in question. In this study, instead of the regular 3-channel aerial photograph, we used a hyperspectral image that carries the surface spectral information on a pixel-by-pixel basis which enables better classification than using only the visible colors. For this purpose, we used two parameters: *SAM* classifier (Eq. (1)) and spectral gradient (m) (Eq. (2)). *SAM* determines the spectral similarity between two spectra by calculating the angle (θ) between two spectra and treating them as vectors in a space with dimensionality equal to the number of bands (Kruse et al., 1993). The purpose is to identify spectral changes occur throughout the entire study area and along the soil's core, using a reference spectrum—with reflectance equaling to unity across the desired spectral range. The *SAM* algorithm is given by:

$$SAM_{error} = cos\,(Rr, Rm) \tag{1}$$

where Rr and Rm are the reference and measured spectrum, respectively. For surface classification, *SAM* was applied on the image using the 440–1302 nm range where high SNR exist, however for identifying changes along the profile and subsurface classification, the entire spectral range (400–2450 nm) was used.

Preliminary examination of the spectral database showed a shift in gradient between 615 and 505 nm in the hyperspectral image and 800 and 500 nm in the ASD measurements. There was an albedo shift in some samples, allowing discernment between either soil types or soil horizons. To classify the topsoil and identify the cause of the changes, the spectral gradient, m, was calculated:

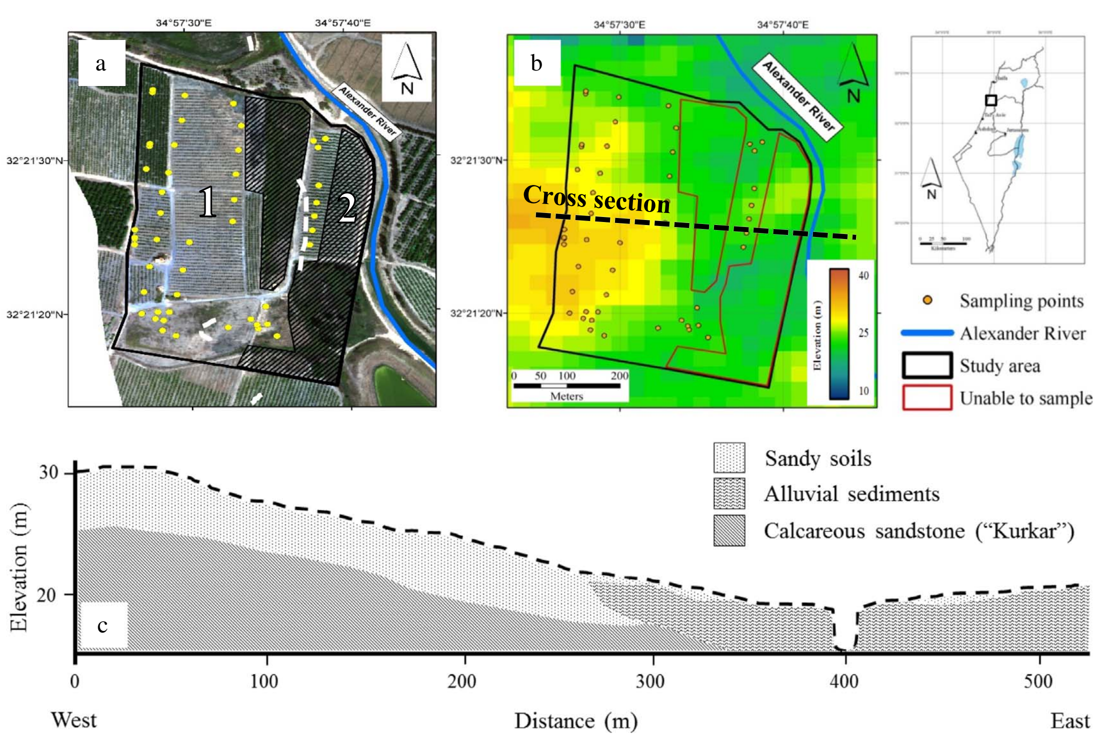


Fig. 1. Aerial photograph of the Ha-Ogen study area and its two primary soil types which were identified in a soil survey (a), elevation data including a cross section (b) and a schematic representation of the cross section which consists of the brown-red sandy soils formed on kurkar in a landscape that includes a riverbed with alluvial sediments (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)
(modified from Singer, 2007).

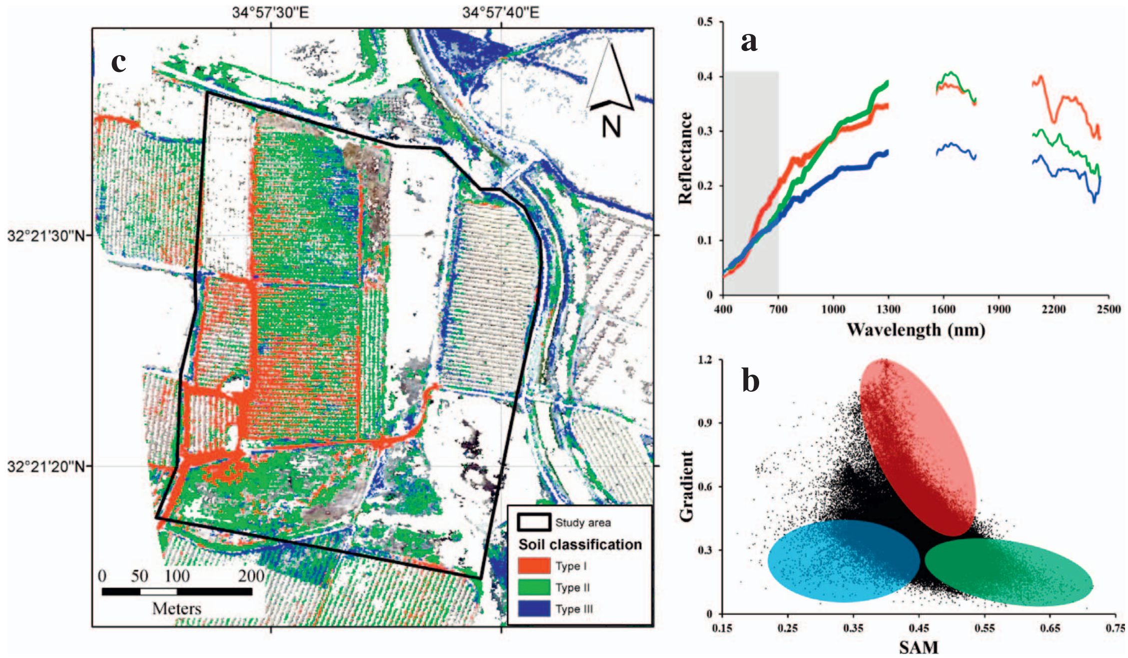


Fig. 2. Surface spectral fingerprints obtained from the FENIX sensor (a), scatter plot of 118,411 surface pixels (b), and the corresponding spatial distribution of the three soil classes in the study area. (c) The map present only pixels which contain soil after applying a masking algorithm on all other surface entities.

$$m = \frac{Ref_2 - Ref_1^*}{\lambda_2 - \lambda_1} 1000 \quad (2)$$

where *Ref* is the reflectance value at a given wavelength and λ is the wavelength in nm (615 and 505 nm in the image data and 800 and 500 nm in the spectral data).

While *SAM* differentiates between sandy soils and alluvial sediments, changes along the sandy soil's cores were observed in the gradient. Therefore, adding the gradient values to the *SAM*–gradient ratio, reveals the changes along the core. The new parameter was determined by:

$$HSC = \left(\frac{SAM}{m}\right) + m \quad (3)$$

where *HSC* is the Hyper-Depth Spectral Classifier Index. In addition, an assessment for the soil type's suitability for plantation has been carried out by identifying gleying conditions which control the subsurface water table, surface drainage and other important soil characteristics. Therefore, the spectral difference (*G*) between 502 nm and 517 nm was calculated (Eq. (4)), which focuses on the iron oxide absorption.

$$G = (CR_{\lambda=502} - CR_{\lambda=517}) * 100 \quad (4)$$

where CR_{λ} is the value of the continuum removal spectrum at the indicated wavelength. Subsequently, we compared our results with subsurface data acquired with a traditional soil survey.

2.6. K-means classification

The K-means method was used for sample clustering (Forgy, 1965). In this method, observations are divided into a given number of clusters by defining the equal number of centroids for each cluster. In our case, each observation which was defined by *SAM*–gradient parameters is classified using the Euclidean distance, as developed by Koolaard and Lawoko (1996), to the nearest centroid. SPSS statistical software (IBM Corp., 2011) was used for this purpose and three clusters were identified.

3. Results and discussion

3.1. Surface classification

While the entire spectral range enables catching the differences between the three spectra, the visible spectral range (marked in gray color) limits this partition for surface types II and III which, in a regular aerial photograph, might look similar as shown in Fig. 2a. This emphasizes the limitation of using only the surface partition of RGB visible images in comparison to hyperspectral technology to generate surface information. Fig. 2b present a scatter plot of *SAM* versus gradient for each soil pixel and the three surface soil types were identified and mapped according to the indices in Table 1. The outcome indicates that the elevated surface in the western parts show high *SAM* and gradient values which is in contrast to the low values obtained in eastern parts, while the surface between the two sections, show high *SAM* accompanied with low gradient. Results are presented in Fig. 2c where the spatial distribution of each soil type is characterized by *SAM*-gradient ratio. The western area, where surface elevation exceeds 26 m, is predominantly surface type I, while the eastern area, closer to the Alexander river where elevation is approximately 20 m, is predominantly surface type III. The middle section of the study area is characterized by a gradual eastern slope where surface type II is defined (see Table 2).

Table 1
The spectral indices of AisaFENIX image for the observed soil types.

Soil type	Indices
I	$\theta_{\lambda=440-1302\ nm} > 0.44$ and $m_{\lambda=615-505\ nm} > 0.45$ and $\theta_{\lambda=440-1302\ nm} > 0.43$ and $m_{\lambda=615-505\ nm} > 0.54$ and $\theta_{\lambda=440-1302\ nm} > 0.42$ and $m_{\lambda=615-505\ nm} > 0.62$ and $\theta_{\lambda=440-1302\ nm} > 0.39$ and $m_{\lambda=615-505\ nm} > 0.70$ and $m_{\lambda=615-505\ nm} > 0.77$
II	$\theta_{\lambda=440-1302\ nm} > 0.44$ and $m_{\lambda=615-505\ nm} < 0.45$
III	$\theta_{\lambda=440-1302\ nm} < 0.44$ and $m_{\lambda=615-505\ nm} < 0.45$

Table 2
SAM and gradient values with their equivalent soil type.

	Low SAM	High SAM
Low gradient	Soil type III	Soil type II
High gradient		Soil type I

3.2. Hyper-depth spectral classification

Due to the effects of frequent alluvial sedimentation and pedoturbation processes, Grumusols do not exhibit any consolidated horizons, resulting in a relatively homogeneous soil profile (Ravikovitch, 1992; Singer, 2007). In contrast, brown-red sandy soils are formed only after sand-stabilization processes occur. Soil formation is initiated when organic matter accumulates and darkens the surface layer, the carbonates leach into the lower strata (Singer, 2007) and the quartz particles redden (rubification) (Ben-Dor et al., 2006; Buol et al., 2011). As a result, this soil type consists of an upper light grayish-brown sandy horizon and a lower reddish-brown or red horizon which gradually turns into yellowish-red loose sand (Singer, 2007). Fig. 3 shows the spectral data of selected cores in Ocean Data View (ODV) software (Schlitzer, 2015), which uses the Data-Interpolating Variational Analysis (DIVA) algorithm (Troupin et al., 2012). The x-axis presents the wavelength in nm, the y-axis is the depth in cm, and the color scale displays the reflectance value. The outcome demonstrates the reflectance value of representative sections, which enables the differentiation of spectral information with depth and the distinction of diagnostic horizons. Cores taken from the southwestern areas of the study area, where brown-red sandy soil predominates, exhibit a significant change in reflectance values with depth. As previously mentioned, in this region, where surface elevation exceeds 26 m, the upper horizon is formed on slight or moderate slopes as a result of roots and other plant residuals that accumulate in the top soil, while carbonates are leached downward (Singer, 2007). In contrast, in the lower regions, as a result of contraction and swelling processes, soils horizons are mixed,

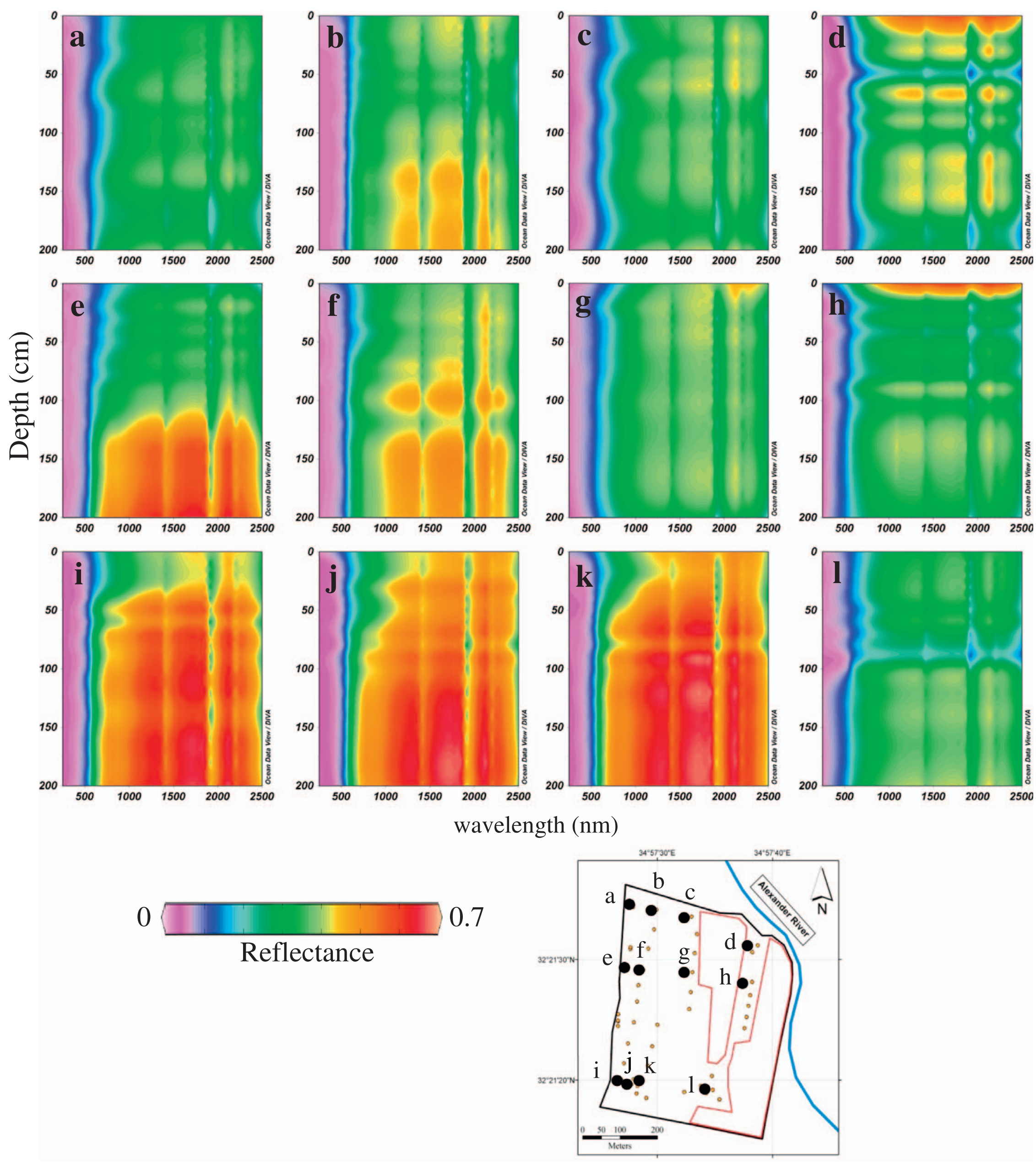


Fig. 3. Spectral data of selected cores (a-l) where reflectance values of each wavelegnth is represented by color.

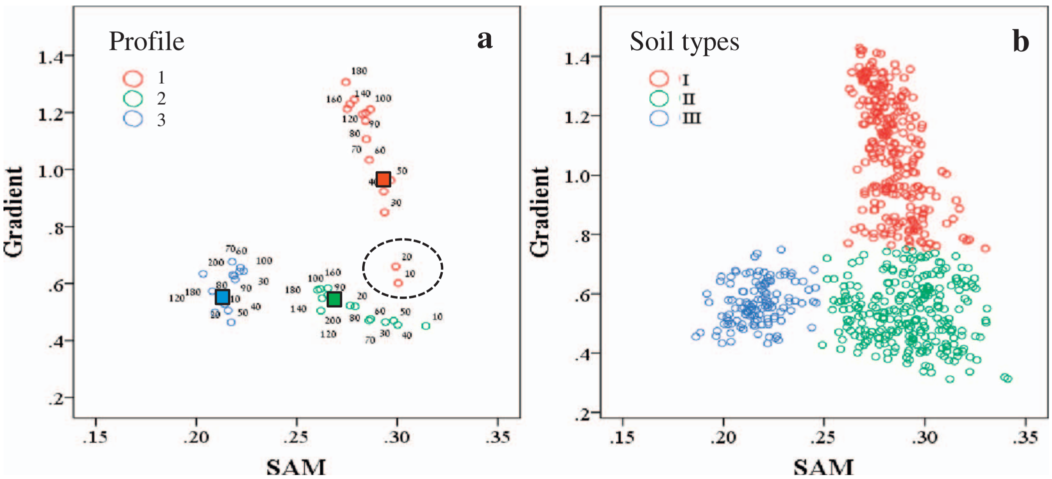


Fig. 4. The three representative profiles taken from the western, central and eastern locations (1, 2 and 3, respectively) together with the calculated centroids (a) and the spectral-based classification using k-means (b). The dashed circle shows the top two samples in profile 1 which were classified as soil type II which the rest of the profile was classified as type I.

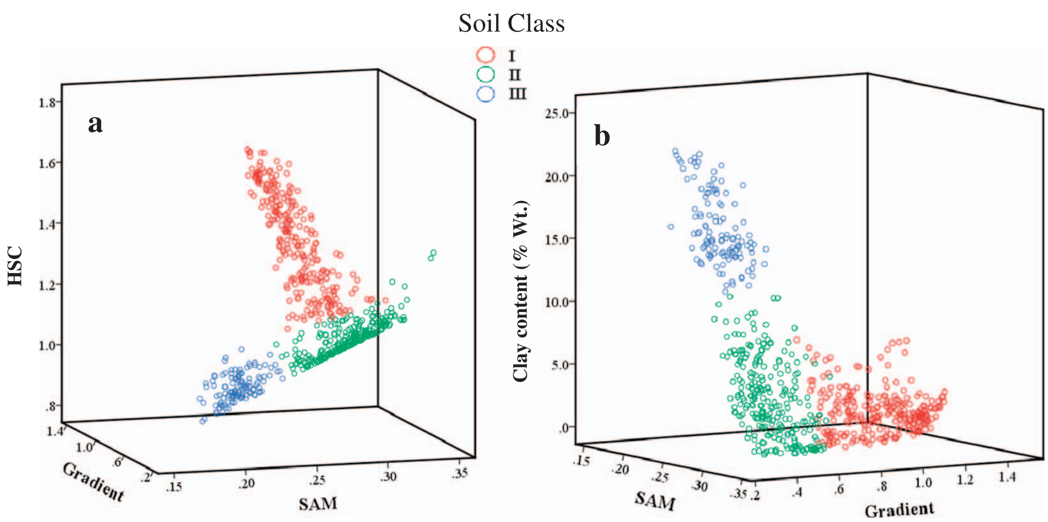


Fig. 5. The spectral-based classification combined with the *HSC* value (a) and clay content (% Wt.) (b) for each soil sample.

resulting in only minor spectral changes along the profile. In these regions, which are under the influence of alluvial sedimentation by the river stream (Ravikovitch, 1992), Grumusols are the predominant soil type.

Analysis of the soil spectral data may improve soil survey by neglecting the color parameter, which is the most traditional and primary classification parameter. While the Spectro-radiometer provides analysis of 2050 wavelengths (in the 400–2450 nm spectral range), the human eye uses a narrow range of wavelengths, which determine the color of an object. While traditional surveys use color as an indicator for transition from one horizon to the next, it is a subjective assessment that varies with the individual.

Spectral-based classification was performed on the core spectral data in a manner similar to the FENIX surface classification. This time, the *SAM* algorithm was applied on the entire spectral range (400–2450 nm) and m was calculated between 800 nm and 500 nm, where the maximum difference along the soil profile is observed. Fig. 4a shows a preliminary analysis of the three profiles which were sampled along the east-west axis in the middle section of the study area which identifies three spectral trends. Hence, the *SAM* and m values of each profile were used for the calculation of the k-means' centroids (square points) using 20 iterations. Subsequently, a scatted plot graph was generated for the entire dataset and the three soil types were identified, as presented in Fig. 4b.

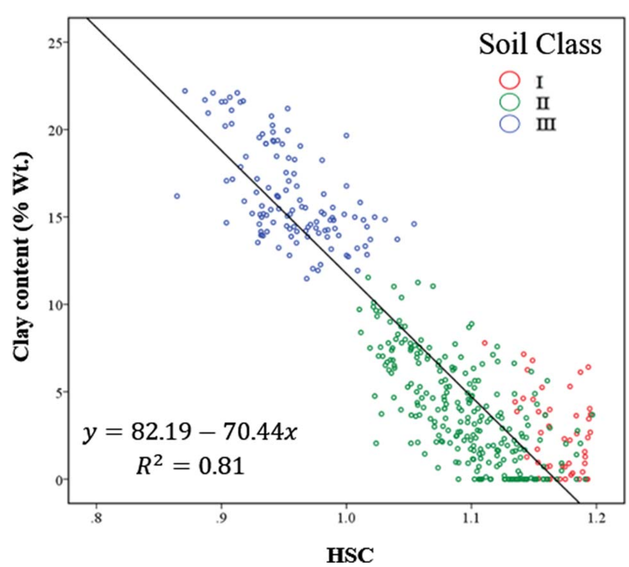


Fig. 6. Correlation between HSC (< 1.2) and clay content in relation to predicted soil class (I, II, III).

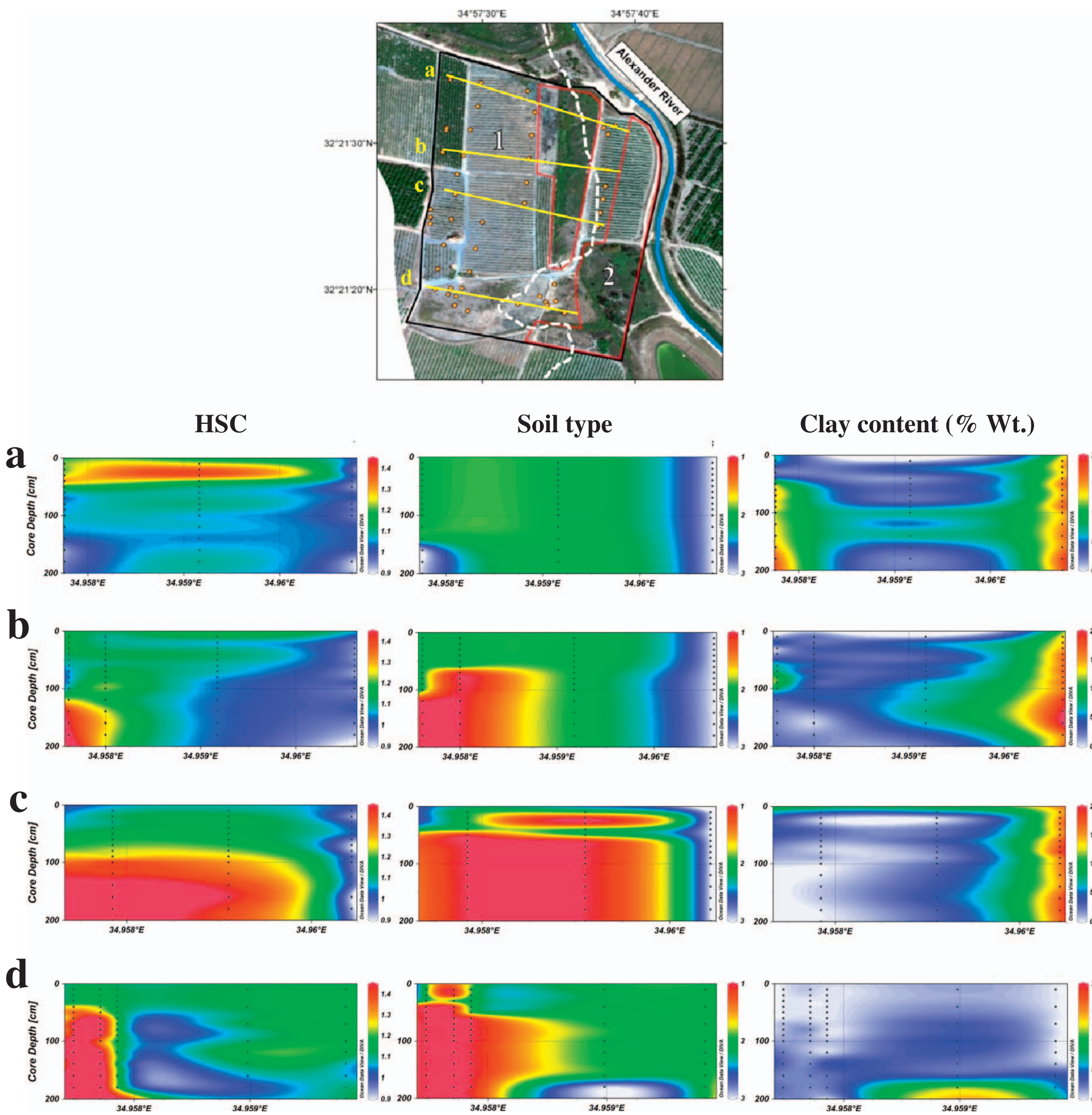


Fig. 7. Cross sections a, b, c, d of the *HSC* index, the predicted soil type and the measured clay content (wt%) in selected profiles.

The soil profiles show three at-depth trends which match the three surface types observed using the FENIX data (Fig. 4b): i) upper horizons with high *SAM* and low gradient values, with increasing gradient with depth (profile 1), ii) upper horizons with high *SAM* and low gradient values, with decreasing *SAM* with depth (profile 2) and iii) low *SAM* and gradient values. The upper two samples in profiles 1 have high *SAM* values accompanied with low gradient values and were classified as soil type II (dashed circle), unlike the deeper samples which were classified as soil type I. The main difference between the profiles appear at depth. While profile 1 displays higher gradient values with depth (from 0.6 to 1.3) and only a slight decrease in *SAM* values (from 0.3 to 0.26), profile 2 presents only a minor increase in gradient values (from 0.5 to 0.6) and a minor decrease in *SAM* values (from 0.31 to 0.26). In contrast, the distribution of samples in profile 3 remains within the limits of both low *SAM* and low gradient values.

Subsequently, the two parameters were plotted together with the HSC as presented in Fig. 5a. Soil type I delivered the highest *HSC* values, followed by types II and III, where the latter gave the lowest. Additionally, due to the fact that soils in this region differ mainly by their texture (The brown-red sandy soil has a sandy to loamy fine sand texture, and the alluvial sediments have a loam to silty loam texture), the clay content (% wt.) was plotted with *SAM*-gradient. Fig. 5b shows soil type I which is characterized by low clay content (mostly lower than 5%), soil type II by 0–12% and soil type III by 12–23% clay content.

Furthermore, the scatter plots in Fig. 5 reveal an important trend between *HSC* index and clay content which is demonstrated in Fig. 6. Low *HSC* values correspond to high clay content and vice versa; however, this correlation existed only when *HSC* values were < 1.2, since at higher *HSC* values, clay content was very low and did not exceed 5%.

Fig. 7 demonstrates the at-depth variation of three parameters: the *HSC* parameter, the predicted soil type of each sample (classified using the k-means algorithm), and the measured clay content, which were plotted in a cross-sectional manner (Fig. 7, cross sections a, b, c, d). The depth (in cm) and the longitudes are shown using the x and y axes, respectively, and the color defines the value of the parameter. The east-west axes of the cross sections were chosen in order emphasize the changes between the dominant soil types.

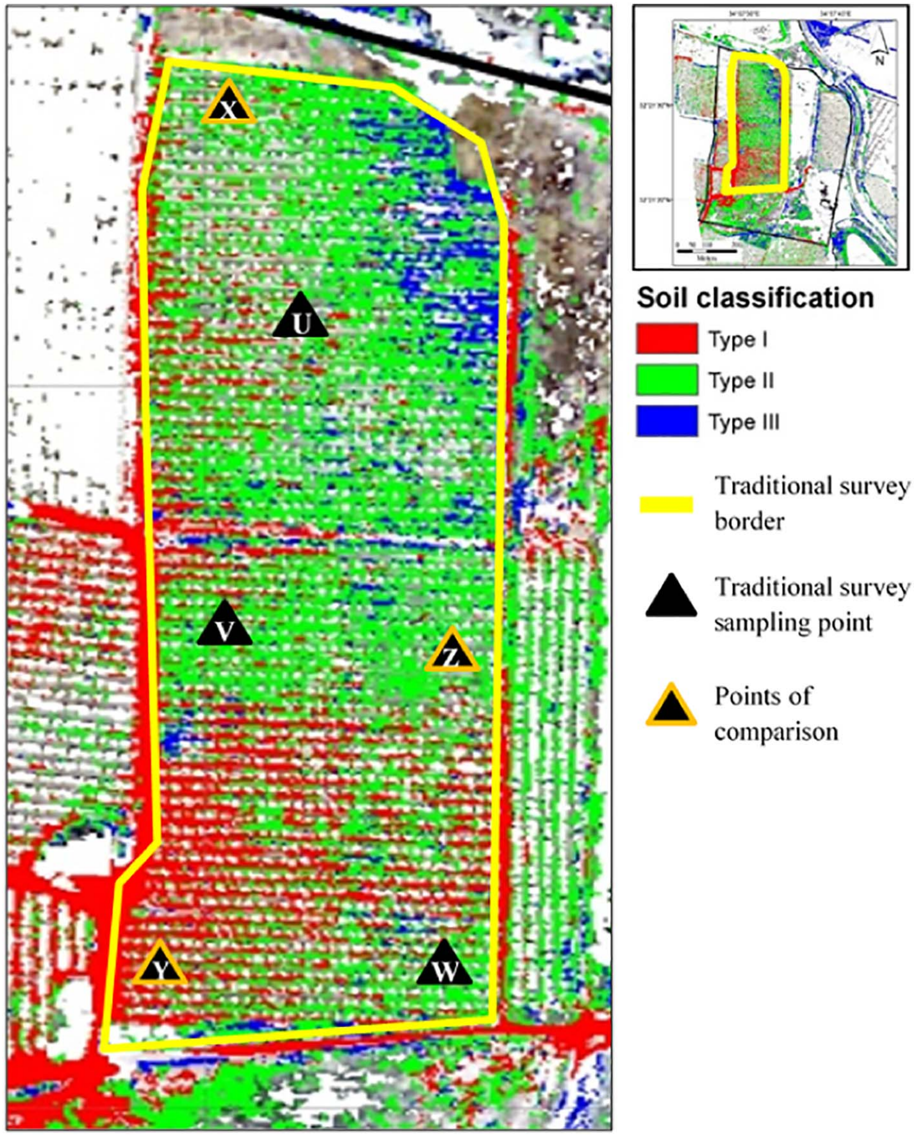
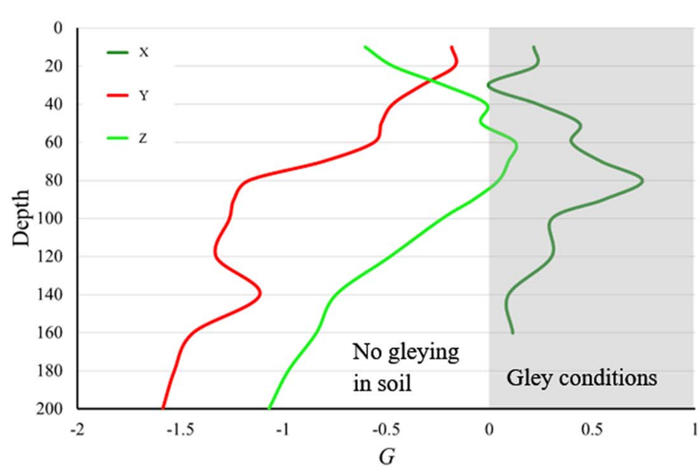


Fig. 8. Location of the six soil profiles presented over the soil types identified by the hyperspectral sensor.



Depth	X	Y	Z
10			
20			
30			
40			
50	**		
60	**		**
70	**		**
80	**		**
90	**		**
100	**		
110	**		
120	**		
130	**		
140	**		
150	**		
160			
170			
180			
190			
200			

Fig. 9. The change in G along the soil profile with a comparison to the presence of gley horizons (marked with **).

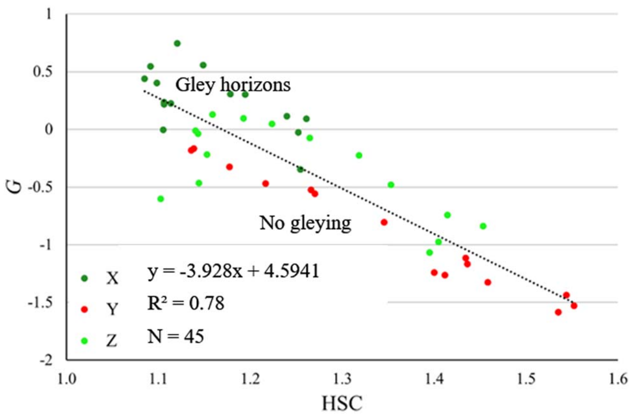


Fig. 10. Correlation between *HSC* and ***G*** parameter in X, Y, Z profiles.

3.3. Comparison with a traditional survey

A comparison between the traditional and spectral-based classifications is essential for evaluating the advantages of the spectral approach. On 19 Apr 2012, a soil survey was conducted in the central region of the study area (the brown-red sandy soil type) and six soil profiles (U–Z) were classified using a pedological description: cumulative alluvial-quartz-loamy-sandy brown Hamra soils with the presence of gley horizons (U, V, W, X and Z) and cumulative alluvial-quartz-loamy sandy brown Hamra without gley horizons, which exhibited good drainage conditions (Y). Fig. 8 presents the locations of the sampling points and the soil type as classified using the hyperspectral sensor.

To validate our results, we performed a comparison with the traditional survey using the soil profiles: X, Y, Z (profiles locations are shown in Fig. 8) which were located near to the ones sampled for this study. Fig. 9 presents the change in *G* value (Eq. (4)) and the presence of gley conditions along the selected profiles (marked with **, as identified by professional soil surveyors from Soil Service Laboratory, Hadera, Israel). Positive *G* values correspond to the presence of gleying conditions while negative values are associated with well-drained soils with no gleying, and a good fit was observed between the two variables.

The *G* parameter was then correlated with the *HSC* index. Low HSC values correspond with high *G* values (gley horizons) and high HSC values correspond with low *G* values (no gleying), resulting an R^2 = 0.78 (Fig. 10). Subsequently, the parameter was applied on the entire spectral database to identify the cores in which gleying conditions exist and locate their distribution in space (Fig. 11).

Results presented herein indicate the ability to classify soil entities using a 3D spectral analysis in both hyperspectral air-borne data and laboratory measurements. In addition, using the HSC parameter resulted in a good assessment of clay content as well as identifying gleying conditions which occur in the subsurface medium.

4. Conclusions

In this work, we demonstrated the efficient, accurate and low-cost use of hyper-depth spectral data for soil classification and soil mapping to optimize future soil surveys. In the last few decades, reflectance spectroscopy has proven to be a precise and rapid tool for quantitative assessment of soil physicochemical properties, mostly in the laboratory. The current study offers a novel spectrally-based method to classify

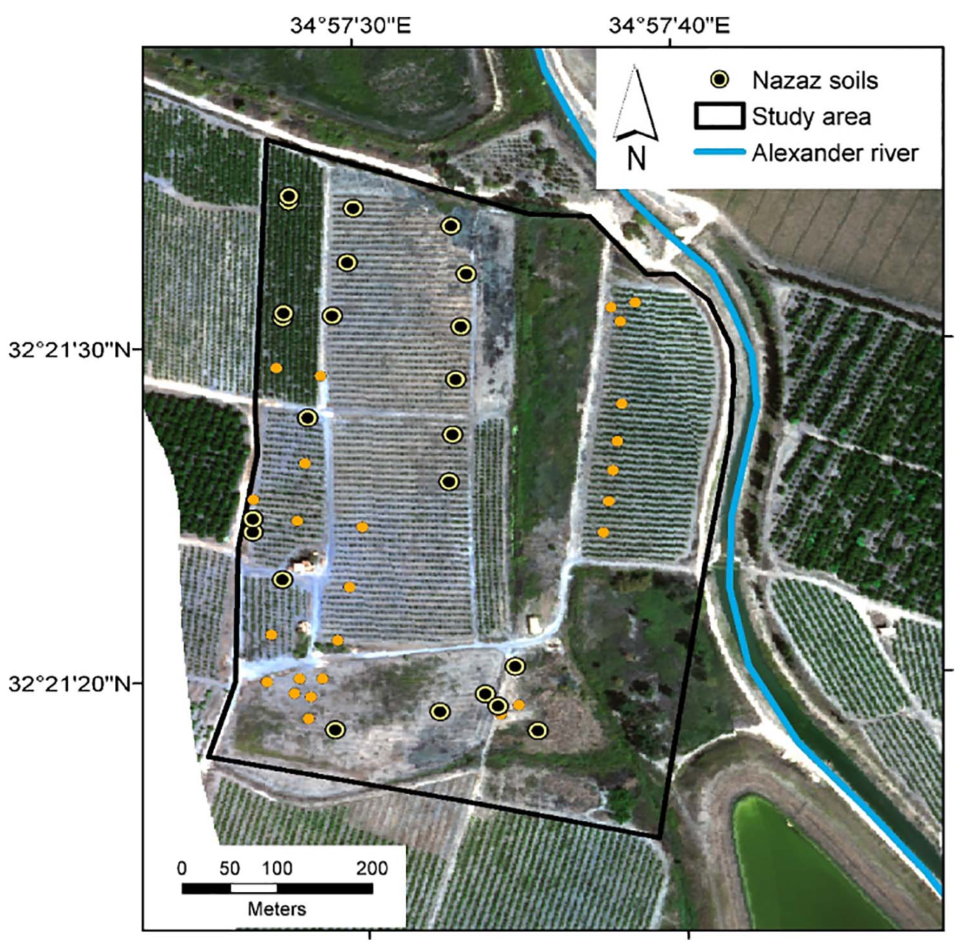


Fig. 11. Spatial distribution of soil profiles which exhibit gleying condition (nazaz soils).

soils using surface and subsurface hyperspectral data. Three primary parameters were used for the classification: the SAM value, which was calculated from a unity reference spectrum, the spectral gradient *m*, and *G* to evaluate the difference between continuum removal values of two wavelengths. The SAM parameter offers insight into the spectral similarity between a reference spectrum and a soil spectrum. High SAM values (above approximately 0.25) correspond to the brown-red sandy soils (Hamra), whereas low SAM values indicate that soils originate from alluvial sedimentation. Contrariwise, the gradient values offer information about the increasing albedo which marks the transition from one horizon to the next along the soil's core. In other words, the SAM threshold determines the soil type, by classifying the samples into two main groups: sandy soils and alluvial soils, and the gradient threshold gives information about changes occur along the core. In addition, the *G* parameter indicates any gleying conditions which may be presence in the soil profile.

As results show, soils in this study area contain both lateral and vertical changes that occur over a short distance due to the physical and chemical characteristics of the catena. The spectral analysis of soil samples indicates the presence of classic catenae in depths where increasing clay content is observed.

The three main conclusions of this research are:

- The combined SAM and gradient values, that are based on reflectance information, can provide important and reliable information about soil-type distribution at the surface level for farmers, researchers and decision-makers.
- The extracted parameter HSC provides satisfactory results for predicting clay content and gley horizons and evaluating catena development, and it is suitable for crops and for 3D classification of soil entities.
- The visual capacity of the human eye is negligible compared to the hyperspectral fingerprint. Converting the hyperspectral data to a 3-wavelength composition is not only unnecessary, it also binds soil surveys to their traditional past.

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